

Servlet Life Cycle

The **Servlet Life Cycle** defines the stages through which a servlet passes from creation to destruction. It is managed by the servlet container.

There are three main methods in the servlet life cycle:

1. init() Method

- Called **only once** when the servlet is loaded for the first time
- Used to perform **initialization tasks**
- Example: loading configuration files, establishing database connections

Key Point: Runs only one time during the servlet's life.

2. service() Method

- Called **every time** a client sends a request
- It processes the request and generates a response
- In HTTP servlets, it calls:
 - doGet() for GET requests
 - doPost() for POST requests

Key Point: Runs multiple times (once per request).

3. destroy() Method

- Called **only once** when the servlet is about to be removed
- Used for **cleanup activities**
- Example: closing database connections, releasing resources

Key Point: Runs once before the servlet is destroyed.

Life Cycle Flow

1. Servlet is loaded into memory
2. init() is called (once)
3. service() is called (for each request)
4. destroy() is called (once before removal)